

A2000 plus LIS Communication Protocol (V0.02) _HL7(Health Level Seven)

Revision control				
Rev.#	Rev. date	Description	Author	Approved by
1	2016/12/15	Initial version	Mingming ZHAI	Jianping HOU
2	2017/8/5	Modification of Lis Protocol in version 170803 of Plus software: 1) Add the NTE segment into instruction 5 to store the batch number of the reagent. Add CE type into the OBX segment to store the positive /negative result of interferon-gamma. Remove the switch of medical record number and completion time from the configuration items. Instead, keep transmitting the medical record number and completion.[set empty in the interferon result if there is a non-interferon item.] 2) Remove the switch of medical record number, instruction 9 is used to transmit medical record number and instruction 10 is used to receive it.	Jie LIU	Zhonghui ZHANG
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1. For readers

This protocol is provided to those who are engaged in the development of the laboratory information system (LIS) and those who need to know about Autobio HL7 interface and is used to guide the LIS developer to develop the LIS interface that makes LIS is able to communicate with the Autobio Automatic Immunoluminescent Analyzer and complete the data transmission successfully. Developers need to have knowledge of LIS,HL7 standards and the capacity of programming skills of network or serial port. The application layer protocol is based on HL7 version 2.3.1. It is suggested that applying including , but not limited to ,Visual C++ and other development tools for the interface development based on Windows platform.

2. This document, HL7 protocol of Autobio Automated Immunoluminescent Analyzer, mainly provides the related content of Autobio HL7 interface protocol including the following two parts:

(1) Autobio HL7 interface.

(2) Communication procedure and message instance.

HL7 is an electronic data exchange standard for inpatient care. It has been adopted by many countries and originally defined by the United States.

The Autobio interface is defined based on HL7 v2.3.1, for more information referring to *HL7 Interface Standards Version 2.3.1*.

Autobio-HL7 interface is the software configurable item of Autobio Automated Immunoluminescent Analyzer. It can adopt TCP/IP or RS232 serial port to realize the LIS communication.

The analyzer asks the LIS system for the sample testing information by scanning the sample bar code in real time. And the test results are reported to the LIS server by real-time and batch transmission while the sample tests are completed.

The several message types, segment types, and other data specified in the HL7 are adopted in this interface protocol because of the small demand of Autobio-HL7 interface.

1.HL7 physical layer

HL7 protocol is based on message as a upper layer protocol and does not provide message start and termination mechanism. In order to determine the message boundary, we use the *Minimal Lower Layer Protocol (MLLP)* protocol in the physical layer.

MLLP protocol is a encapsulation method of HL7 message, which is defined by HL7 standard. Message encapsulation means that the message starts with a single character and ends with two characters.

Message transmission format in communication layer:

<SB> ddddd <EB><CR>

<SB> = Start Block character (1 byte)

<VT> (vertical table), of which ASCII code is 0x0B, is distinguished with the character SOH or STX.

dddd = data (variable number of byte)

“**dddd**” is a HL7 message that contains only ISO 8859 -1 characters (hexadecimal value from 20 to FF) and <CR>, excluding characters that cannot be printed and other control characters. It is also the responsibility of the external system not to send any prohibited characters, nor will Autobio-HL7 interface respond for these characters.

<EB> = End block character (1 byte)

<FS>, of which ASCII code is 0x1C, is distinguished with the character ETX or EOT.

<CR> = Carriage return (1 byte)

Of which ASCII code is 0x0D

2.Nonconformance of HL7 standard

The HL7 interface is specially customized according to the actual business needs reusing some message segments or fields, which does not fully conform to the standard of HL7 v2.3.1.

3.General syntax of HL7 interface

In HL7 communication protocol, message is the basic unit of data exchange between systems, each message has its own message types.

A message consists of multiple segments with names that define its content or functionality.

A segment is composed of multiple data fields.

The first segment of a message is always a message head segment, which indicates the program name, message type, and a unique message ID number to be sent and received. The composition of the next segment is determined by the type of message. For example, PID segment (Patient Identification Data) includes name, address, social security number, etc.

A data field may also consist of multiple components.

For instance:

MSH|^~\&|A2000 Plus|Autolumo|||20161027194802||ORU^R01|1|P|2.3.1|||0|ASCII|||

Here: The five character definitions follow MSH are used to distinguish the delimiters of fields, components, and subcomponents. The Autobio-HL7 standard uses the following characters:

Character	Description
	Field delimiter
^	Component delimiter
&	Subcomponent delimiter
~	Repeated delimiter
\	Escaped delimiter

The first field of the MSH includes kinds of delimiters. Some of the later fields are empty because they are optional and the Autobio-HL7 interface does not use it. The detailed definition and selection of the fields will be explained in detail in the specific message. For any message, the sequence of the segments after the MSH segment is specified. The following sections describe the sequence in detail, and use the following syntax structure to indicate the selection or repetition of the segment:

[] indicates the selection of the segment inside.

{ } indicates that the paragraphs can be repeated 0 or 1 or more times.

4. Available message of HL7

The available messages of HL7 in Autobio-HL7 are QRY/DSR and ORU/ACK.

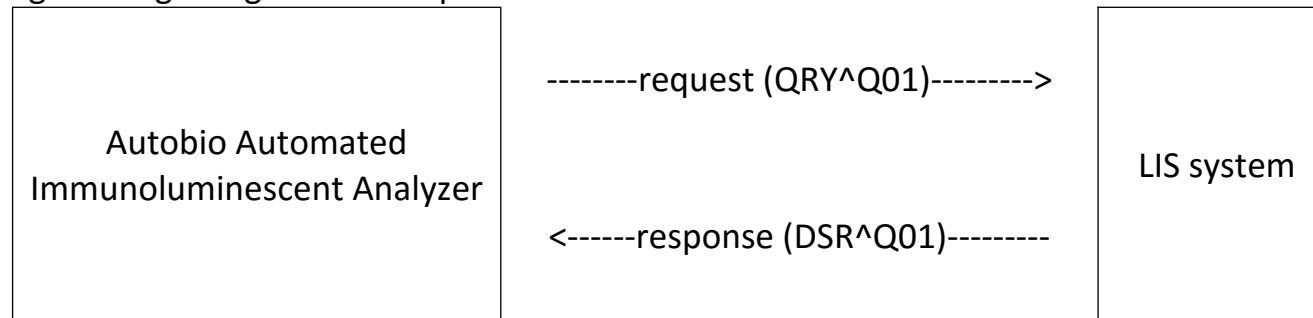
Here :

QRY/DSR used as an Autobio Analyzer to request the LIS to obtain a sample test request.

ORU/ACK used as an Autobio Analyzer to send the test results to the LIS.

4.1 Get the test request(QRY/DSR)

The diagram of getting the test request is as follows:



HL7 message

QRY/DSR: initial mode status display query – response immediately

QRY/DSR messages is used to query the LIS system for sample request information, which triggers event Q01. We use it to request a patient sample test from the LIS system.

Concrete structures are as follows:

Request:

ORU	Observational Results
MSH	Message header
PID	Patient information data required by network(patient name, sample number, etc.)
OBR	Doctor's advice (sample type, transfer doctor, examiner, clinical diagnosis, etc.)
{OBX}	Result

Response:

ACK	Acknowledgement
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MSH	Message header
MSA	Message acknowledgement

5.Message segment instruction of HL7

5.1.1. MSH – Message Header

MSH is the first message segment of HL7 message, and all messages begin with a MSH segment, usually at the top of the message.

This message segment is used to define the intention, source, destination, and some details of the message syntax.

MSH segment of Autobio-HL7 interface adopts the fields as follows:

Ordinal	Segment name	Length	Description
1#	Field Separator	1	Contains a delimiter between the segment ID and the first real field, defining the field delimiter for the rest of the message, here is “ ”.
2#	Encoding Characters	4	Contains component delimiters, repeating delimiters, escape delimiters, and subcomponent delimiters, here are “^~\&”.
3	Sending Application	180	Sending application, here is Autobio.
4	Sending Facility	180	Sending facility, here is the model number, for example, AutoLumo A2000 Plus
7	Date/time of message	26	Date/time of current message. Here is the system date/time, for instance, 20170630093523
9#	Message type	7	Message type, for example, ORU^R01
10#	Message control ID	20	Message control ID, used to command words, ※2
11#	Processing ID	3	Processing ID, here is “P”(indicate product)
12#	Version ID	60	Version ID, HL7 protocol version, here is 2.3.1

13	Sequence number	15	Sequence number, here is the current time, the format is "yyMMddHHmmssfff", used to be a running number. For instance : 170630093523613
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※1:

The field that must be included in the HL7 segment with the symbol "#" in the ordinal number.

※2:

The command words used in the Autobio-HL7 protocol are as follows:

Command words	Sender	Receptor	Description
1	A2000 Pus	LIS	Get the sample detailed test request data for the specified sample number and the specified medical record number.
	LIS	A2000 Pus	The response gets the sample detail test request data for the specified sample number and the specified medical record number.
3	A2000 Pus	LIS	Gets the number of all patient test samples assigned to the designated instrument.
	LIS	A2000 Pus	Gets the response of the LIS assigned to all patient test samples for the specified instrument.
5	A2000 Pus	LIS	Send test results to LIS by test.
	LIS	A2000 Pus	Send test results to LIS response by test
7	A2000 Pus	LIS	Send test results to LIS by sample.
	LIS	A2000 Pus	Send test results to LIS response by sample.
9	A2000 Pus	LIS	Quickly obtain sample detail test request data for a specified sample number and a specified medical record number.
	LIS	A2000 Pus	The response quickly gets the sample detail test request data for the

			specified sample number and the specified medical record number.
15	A2000 Pus	LIS	Quickly get the specified sample frame number test request data.
	LIS	A2000 Pus	Response quickly gets the specified sample frame number for detailed test request data.

5.1.2. QRD – Query Definition

QRD segment is defined as a query, and each QRY message contains this segment, LIS needs to return the initial QRD segment while responding.

The QRD segment of the Autobio-HL7 interface uses the following fields:

Ordinal (※1)	Segment name	Length	Description
1#	Query date/time	26	The current date/time, here is the system date/time, of which format is “yyyyMMddHHmmss”, for instance, 20170630093523
2#	Query format code	1	Query format code, here is “R”(record – oriented format)
3#	Query priority	1	Query priority, here is “D”(deferred)
4#	Query ID	10	Query ID indicates different query requests, here is system date/time, of which format is “HHmmssfff”, for example, 093523614
7#	Quantity limited request	10	Quantity limited request, here is “RD”.
8#	Who subject filter	60	Who subject filter, which is used as the sample number in obtaining the test according to the sample number, and in obtaining the test according to the sample racks and position;
9#	What subject filter	60	What subject filter, here is “OTH”.(other)

10#	What department data code	60	What department data code. When obtaining a test by sample racks and sample position, it is used as the current sample position(1 to 5), the others set empty.
12	Query result level	1	Query result level, here is "T".(full results)

※1:

The field that must be included in the HL7 segment with the symbol "#" in the ordinal number.

5.1.3. QRF – query filter

The meaning of QRF segment is query filter, QRY message can be selected, if the request sent QRF segment, the receiver should return the QRF segment as initial status.

The QRF segment of the Autobio-HL7 interface uses the following fields:

Ordinal(※1)	Segment name	Length	Description
1#	Where subject filter	20	Here is the instrument model, for instance, AutoLumo A2000 Plus
2	When data start date/time	26	Here is the system date and the time 00: 00: 00. The format is "yyyyMMddHHmmss".
3	When date end date/time	26	Here is the system date/time. The format is "yyyyMMddHHmmss".
6	Which date/time qualifier	12	Here is "RCT".(Specimen receipt date/time, receipt of specimen in filling ancillary(Lab))
7	Which date/time status qualifier	12	Here is "COR".(Corrected only (no final with corrections))
8	Date/time selection qualifier	12	Here is "ALL".(all values within the range)

※1:

The field that must be included in the HL7 segment with the symbol "#" in the ordinal number.

5.1.4. DSC – continuation pointer

The DSC segment is used to indicate whether the response message is the last data message.

The DSC segment is not currently used in the Autobio-HL7 interface message.

5.1.5. MSA – query filter

The MSA message segment is used to query responses and acknowledgements. It can appear in the DSR^Q01 and ACK^R01 message. Field 4 and field 6 use integer data type; the rest of the fields use string data type.

The MSA segment of the Autobio-HL7 interface uses the following fields:

Ordinal (1)	Segment name	Length	Description
1#	Acknowledgement code	2	“AA” for acceptance and “AE” for error.
2#	Message control ID	20	Same as the sender MSH-10 field, used as a command word, please refer to the note description of the 5.1.1MSH segment for detailed command word information
3	Text message	80	Text message, when occurring error or rejection, a text description of an event. Corresponds to field 6th. Used to write the error log.※2
4	Expected sequence number	15	Expected sequence number※3
6	Error condition	100	Error code, 0 for success or else for error

※1

The field that must be included in the HL7 segment with the symbol "#" in the ordinal number.

※2

When sending test result response message in ACK^R01, the method of sending test result is different, and the value of field is also different. The mode of sending test results is determined by the command word field (MSH-10) in the MSH segment of the sender message. Please refer to the 5.1.1MSH section remarks for detailed command word information. Specific settings are as follows:

(1)When sending test result response according to the test (command word: 5): [sample bar code, patient medical record number, test item number], for example: 1,Tom,113.

Sample bar codes can be obtained from the OBR segment sent by the requester, as described in 5.1.10 OBR.

Patient medical record numbers can be obtained from the PID segment sent by the requesting, as described in paragraph 5.1.9 PID.

(2) When sending test result response according to sample (command word: 7): [sample bar code, patient medical record number], for example: 1,Tom.

Sample bar codes can be obtained from the OBR segment sent by the requester, as described in 5.1.10 OBR.

The patient medical record number can be obtained from the PID segment sent by the requesting, or, if there is no PID segment, set null, as described in paragraph 5.1.9 PID.

※3

When sending test result response message in ACK^R01, the method of sending test result is different, and the value of field is also different. The mode of sending test results is determined by the command word field (MSH-10) in the MSH segment of the sender message. Please refer to the 5.1.1MSH section remarks for detailed command word information. Specific settings are as follows:

(1) when sending a test result response by test (command word: 5): read the test request ID set by the requestor in the OBX segment of the send test result request ORU^R01 message

(2) when sending the test result response by sample (command word: 7): read the sample bar code set by the requester in the OBR segment of the sending test result request ORU^R01 message.

5.1.6. ERR – error

The ERR segment is used to add an error description to the confirmation message, which is used in the DSR^Q01 message in the Autobio-HL7 interface. The message segment has only one field, using integer data type.

The ERR segment of the Autobio-HL7 interface uses the following fields:

Ordinal (※1)	Segment name	Length	Description
1#	Error code and location	80	0 for success or else for error

※1

The field that must be included in the HL7 segment with the symbol "#" in the ordinal number.

5.1.7. QAK – query acknowledgement

The QAK segment contains information that follows the query response, which appears in the DSR^Q01 message. All fields use string data types.

The QAK segment of the Autobio-HL7 interface uses the following fields:

Ordinal(※1)	Segment name	Length	Description
1	Query tag	32	Here is "SR".(samples request information)
2	Query response status	2	OK means found data and no error; NF found no data, no error; AE means program error; AR means program refused.

※1

The field that must be included in the HL7 segment with the symbol "#" in the ordinal number.

5.1.8. DSP – display data

The DSP segment is used by the LIS system to return the sample request information and patient information obtained from the query to the requester. The DSP message segment can be repeated, and the message segment will appear in the DSR^Q01 message.

The first field uses integer data types. Other fields use string data types. The MSA segment of the Autobio-HL7 interface uses the following fields:

Ordinal(※1)	Segment name	Length	Description
1	Set ID – DSP	4	Confirm different DSP segment ※2
3#	Data line	300	Contents queried※3

※1

The field that must be included in the HL7 segment with the symbol "#" in the ordinal number.

※2, ※3:

When used in DSR^Q01 to get the response message of the test request, the method of obtaining the test is different, and the value of the [1] segment is different from that of the [3#] segment. The mode of the test request is determined by the command word field (MSH-10) in the MSH segment of the sender message.

Please refer to the note description of the MSH segment for detailed command word information. Specific settings are in the following table:

Get test mode	Command word	Numbers of DSP segment	DSP value	
			Set ID – DSP	Data line
Get the sample detailed test request data for the specified sample number and the specified medical record number	1	2*n+15	1	Dilution factor (multiple of samples diluted)
			2	STAT 0:NO 1:YES
			3	Medical record number
			4	Last name
			5	First name
			6	Gender 0:female 1:male

			7	Age
			8	Apartment
			9	Ward
			10	Bed number
			11	Submitting staff
			12	Submitted date (YYYY/MM/DD)
			13	Examiner
			14	Note
			15	Number of test item
			16	Item number of the first test
			17	Dilution factors of the first test (the factors diluted on the instrument must be within the scope of support for the test item)
			18	Item number of the second test
			19	Dilution factors of the second test (the factors diluted on the instrument must be within the scope of support for the test item)
			...	...
			2*N+14	Item number of the Nth test
			2*N+15	Dilution factors of the Nth test (the factors diluted on the instrument must be within the scope of support for the test item)
Gets all patient tests assigned to the designated	3	2*N+1	1	Number of test samples assigned to the instrument (N)
			2	The first sample number

instrument(not use in this protocol)			3	The first sample patient medical record number
			4	The second sample number
			5	The second sample patient medical record number
			...	...
			2*N	The Nth sample number
			2*N+1	The Nth sample patient medical record number
Quickly obtain sample details of a specified sample number and a specified medical record number. Please ask for data	9	2*N+5	1	Dilution factor (multiple of samples diluted)
			2	STAT 0-NO 1-YES
			3	Sample type (Type that can be recognized by A2000 plus, if not, regard as the serum)
			4	Patient medical record number
			5	Test item number(how many test in one sample)
			6	Item number of the first test
			7	Dilution multiple of the first test (diluted on the machine must be within the scope of support for the test project)
			8	Item number of the second test
			9	Dilution multiple of the second test (diluted on the machine must be within the scope of support for the test project)
			...	...
			2*N+4	Item number of the Nth test
			2*N+5	Dilution multiple of the Nth test (diluted on the machine must be within the scope of support

				for the test project)
Quickly obtain detailed test request data for specified frame numbers	15	$2*N+9$	1	Sample racks number
			2	Sample position
			3	Sample status 1-exist 0-null
			4	Sample number
			5	Dilution factor (multiple of samples diluted)
			6	STAT 0-NO 1-YES
			7	Patient medical record number
			8	Sample type (Type that can be recognized by A2000 plus, if not, regard as the serum)
			9	Test item number(how many test in one sample)
			10	Item number of the first test
			11	Dilution multiple of the first test (diluted on the machine must be within the scope of support for the test project)
			12	Item number of the second test
			13	Dilution multiple of the second test (diluted on the machine must be within the scope of support for the test project)
			...	
			$2*N+8$	Item number of the Nth test
			$2*N+9$	Dilution multiple of the Nth test (diluted on the machine must be within the scope of support for the test project)

5.1.9. PID – patient identification

The PID segment is mainly used to transmit the patient information. This message will appear in the ORU^R01 message. When the medical record number switch in the A2000 Plus instrument client system is turned on, the Autobio-HL7 interface uses this segment to send the test sample patient medical record number information to the LIS server. After receiving the test results, the LIS server returns the patient case number in the MSA segment of the response message in the specified format.

The PID segment of the Autobio-HL7 interface uses the following fields:

Ordinal(※1)	Segment name	Length	Description
3#	Patient identifier list	20	Patient medical record number
5#	Patient name	48	Patient name, reserved, set null

※1

The field that must be included in the HL7 segment with the symbol "#" in the ordinal number.

5.1.10. OBR – patient identification

The OBR segment is used to transmit information about the doctor's orders related to the test report. This message will appear in the ORU ^ R01 message. The Autobio-HL7 interface uses this segment to send the test sample bar code, sample ID and other information to the LIS server. After receiving the test results, the LIS server returns the patient sample bar code in the MSA segment of the response message in the specified format.

The OBR segment of the Autobio-HL7 interface uses the following fields:

Ordinal(※1)	Segment name	Length	Description
1	Set ID-OBR	10	Here is "1"
2	Placer order number	22	Used as sample bar code

3	Filler number	22	Used as sample ID
4#	Universal service ID	200	Used as mark of instrument , for instance , AutoLumo A2000 Plus

※1

The field that must be included in the HL7 segment with the symbol "#" in the ordinal number.

5.1.11. OBX – patient identification

OBX is primarily used to pass observation information in a report message. The message appears in the ORU ^ R01 message, which is used by the Autobio-HL7 interface to send the test result information to the LIS server. A patient sample may have multiple OBX, interfaces that do not provide reproducibility testing for test data, which are verified by LIS systems.

The OBX segment of the Autobio-HL7 interface uses the following fields:

Ordinal(※1)	Segment name	Length	Description
1	Set ID – OBX	10	Other empty as test request ID; when sending results by test
2	Value Type	3	Here is “CE”
3#	Observation Identifier	590	Used as test item number
4#	Observation Sub-ID	20	Used as test item number
5	Observation Value	65536	Used as [RLU]^[concentration or s/co]
11#	Observe Result Status	1	Here is “F”
14	Date/Time of the Observation	28	Used as completion time

※1

The field that must be included in the HL7 segment with the symbol "#" in the ordinal number.

5.1.12.NTE - Patient Identification

NTE is mainly used to pass comments or instructions in a report message. The message appears in the ORU ^ R01 message, which is used by the Autobio-HL7 interface to send the batch number information of the test reagent to the LIS server.

The NTE segment of the Autobio-HL7 interface uses the following fields:

Ordinal(※1)	Segment name	Length	Description
3#	Comment	65536	Used as lot number of reagent

※1

The field that must be included in the HL7 segment with the symbol "#" in the ordinal number.

Communication procedures and message examples:

Communication type	C o m m a n d w o r d	Send message demo	Response message demo
Obtain sample detailed test request data with a specified	1	MSH ^~\& Autobio AutoLumo A2000 Plus 201703061	MSH ^~\& Autobio LIS 20170306120215 DSR^Q01 1 P 2.3.1 17030612032 5979 MSA AA 1 ERR ^^^0

sample number and a specified medical record number		20325 QRY^Q01 1 P 2.3.1 170306120325979 QRD 20170306120325 R D 120325982 RD 14081916 OTH T QRF AutoLumo A2000 Plus 20170306000000 20170306120325 RCT COR ALL	QAK SR OK QRD 20170306120325 R D 120325982 RD 14081916 OTH T QRF AutoLumo A2000 Plus 20170306000000 20170306120325 RCT COR ALL DSP 1 1 DSP 2 0 DSP 3 serum DSP 4 DSP 5 DSP 6 0 DSP 7 20 DSP 8 apartment DSP 9 ward DSP 10 bed DSP 11 DSP 12 2017/2/17 11:09:05 DSP 13 name DSP 14 null DSP 15 2 DSP 16 159 DSP 17 1
Get patient tests assigned to specified instruments (not use in protocol)	3		

Send test results to LIS by test	5	MSH ^~\& Autobio AutoLumo A2000 Plus 20170628153005 ORU^R01 5 P 2.3.1 170628153005419 PID BL20140226 OBR 1 2017062703 211 AutoLumo A2000 Plus NTE 180518 OBX 387 CE 121 121 14^0.1^N F 2017/06/28 15:30:05	MSH ^~\& Autobio LIS 20170628152923 ACK 5 P 2.3.1 170628153005419 MSA AA 5 2017062703,BL20140226,121 387 0
Send test results to LIS by sample	7	MSH ^~\& Autobio AutoLumo A2000 Plus 20170629165510 ORU^R01 7 P 2.3.1 170629165510846 OBR 1 2017062701 341 AutoLumo A2000 Plus	MSH ^~\& Autobio LIS 20170629165428 ACK 7 P 2.3.1 170629165510846 MSA AA 7 2017062701, 341 0

		OBX CE 150 0^ 0 F OBX CE 151 15 000000^1118.35 F OBX CE 152 0^ 0 F OBX CE 153 9^ 0 F OBX CE 154 0^ 0 F OBX CE 155 12 ^0 F OBX CE 156 0^ 0 F OBX CE 157 0^ 0 F OBX CE 158 5^ 0 F OBX CE 159 23 ^0 F	
Quickly obtain sample detail test request data for a specified	9	MSH ^~\& Autobio AutoLumo A2000 Plus 201706291 43426 QRY^Q01	MSH ^~\& Autobio LIS 20170629143343 DSR^Q01 9 P 2.3.1 17062914342 6340 MSA AA 9 ERR ^^^0 QAK SR OK

sample number and a specified medical record number		9 P 2.3.1 170629143426340 QRD 20170629143426 R D 143426340 RD 2017062725 OTH T QRF AutoLumo A2000 Plus 20170629000000 20170629143426 RCT COR ALL	QRD 20170629143426 R D 143426340 RD 2017062725 OTH T QRF AutoLumo A2000 Plus 20170629000000 20170629143426 RCT COR ALL DSP 1 1 DSP 2 0 DSP 3 serum DSP 4 BL25 DSP 5 1 DSP 6 167 DSP 7 1
Quickly obtain detailed test request data for specified sample holder number	15	MSH ^~\& Autobio AutoLumo A2000 Plus 20161207100639 QRY^Q01 15 P 2.3.1 161207100639800 QRD 20161207100639 R D 100639802 RD DA56 OTH 1 T QRF AutoLumo A2000 Plus 20161207000000 20161207100639 RCT COR ALL	MSH ^~\& Autobio LIS 20161207100833 DSR^Q01 15 P 2.3.1 161207100639800 MSA AA 15 ERR ^^^0 QAK SR OK QRD 20161207100639 R D 100639802 RD DA56 OTH 1 T QRF AutoLumo A2000 Plus 20161207000000 20161207100639 RCT COR ALL DSP 1 1 DSP 2 14081901 DSP 3 1 DSP 4 0 DSP 5 BL20140226 DSP 6 serum DSP 7 2

		000 20161207100 639 RCT COR A LL	DSP 8 122 DSP 9 1 DSP 10 100 DSP 11 1
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